



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Course: CSE489

Name: Fahim Montasir

ID: 23141013

Email: fahim.montasir1@g.bracu.ac.bd

Project Title: MoveCrew (Smart Logistics & Workforce Tracking System)

Project Features:

By the end of this course, implemented features:

a) Role-based login and authentication (Admin, Mover, Customer) via Supabase Auth, with distinct navigation flows and dashboard views for each user type.

b) Customer Job Request System — a comprehensive interface for customers to create move requests with pickup/destination details, move dates, and specific instructions.

c) Itemized Inventory Management — granular tracking of items for each job (quantity and status), allowing for progress updates like "Packed", "Loaded", and "Unloaded".

d) Real-time GPS Tracking — background capture of mover locations during active assignments, allowing for precise monitoring of the workforce.

e) Workforce Attendance Logging — a dedicated "Clock-in/Clock-out" system for movers that records timestamps and statuses tied directly to specific job assignments.

f) Admin Assignment Engine — a powerful dashboard for Admins to review approved jobs and assign multiple movers based on availability and employee codes.

g) Mover Task Management — an interactive list for movers to view assigned tasks, respond to job offers (Accept/Decline), and manage their active work queue.

h) Customer Feedback & Reviews — a post-job completion feature where customers can rate their movers and provide detailed comments on the service.

i) Automated Status Workflow — a robust state-machine logic (using PostgreSQL and Flutter) that manages job transitions from Pending to In-Progress and finally to Completed.

j) Supabase Integration — full utilization of a relational database schema with foreign key constraints between Users, Movers, Jobs, and Assignments.

Online Resources used:

a) Reference:

- *Flutter official documentation (docs.flutter.dev)*
- *Supabase official documentation (supabase.com/docs) — Auth, Database, and Real-time listeners*
- *Google Maps for Flutter documentation for location-based services*

Future Enhancements:

Following enhancements can be added to the current system which will improve the system:

a) Internal Messaging System — implementing real-time chat between Customer-Mover and Mover-Admin for better coordination during the moving process.

b) iOS platform support — currently optimized for Android; future development will include full iOS compatibility and UI refinement.

c) Push Notifications — integration of Firebase Cloud Messaging (FCM) to alert movers instantly when a new job is assigned to them.

d) Route Optimization — using map APIs to calculate the most efficient path for movers between pickup and destination points.

e) In-App Payment Gateway — allowing customers to securely pay for their move via digital wallets or cards upon job completion.